Grade 7 Accelerated
Unit 3
Lesson 8 Practice

Name___Answer Key_______
Date______________________
Period______________________

Use a calculator to complete the table. Find the value of each mathematical statement. Then, if possible, rewrite the value as a power of 10.

Mathematical Statement	Value Found Using the Calculator	Value Written as a Power of 10
1. $\frac{10^5}{10^{12}}$	0.0000001	10^{-7}
2. $10^6 + 10^4$	101000	Not Possible
3. $10^7 \times 10^2$	$1{,}000{,}000{,}000$	10^9

4. Which of the mathematical statements could be rewritten using a power of 10? Explain your reasoning.

We can rewrite $\frac{10^5}{10^{12}}$ and $10^7 \times 10^2$ as powers of 10.
In $\frac{10^5}{10^{12}}$, we are dividing like bases, so we can use the quotient of powers law for exponents. In $10^7 \times 10^2$, we are multiplying like bases, so we can use the product of powers law for exponents.

5. Which of the mathematical statements could not be rewritten using a power of 10? Explain your reasoning.

$10^6 + 10^4$ cannot be written as a power of ten since we are adding, not multiplying, and the bases do not have the same exponent.

6. Complete the table.

Scientific Notation	Standard Form
4.1×10^5	$410{,}000$
8.2×10^6	$8{,}200{,}000$
1.64×10^7	$16{,}400{,}000$

7. How many times larger is 8.2 than 4.1?

2

8. How many times larger is 10^7 than 10^5?

100

9. How many times larger is 8.2×10^7 than 4.1×10^5?

200

10. The population of visitors going to Disney World was about 6.94×10^6 in 2020. The population of visitors going to Universal Studios in 2020 was about 1.7×10^6.

Complete the statement.

In 2020, the population of visitors to Disney World was about ____4____ times larger than the population of visitors going to Universal Studios.

$\frac{6.94}{1.7} \times \frac{10^6}{10^6} \approx 4 \times 1$